Kompetenzen

CORE JAVA

SPRING BOOT

FULL STACK

REACT

AI INTEGRATION

CAREER

KOMPETENZEN TECHNOLOGIES

Full Stack Java Developer

135-Day Project-Driven Career Program

270 Technical Hours · 12 Phases

135 DAYS **270** TECHNICAL HOURS **02** MAIN PROJECTS

> Engineer. Build. Integrate.

Kompetenzen Technologies

PROGRAMME OVERVIEW

Programme at a glance

A project-driven developer programme that takes you from Java fundamentals to building secure Spring Boot and React applications — with AI built into how you work.

135	270	2	Project-driven	02
DAYS	TECHNICAL HOURS	HOURS PER DAY	LEARNING MODEL	MAIN PROJECTS

WHAT THIS PROGRAMME SETS OUT TO DO

01 Learn to program in Java.

02 Learn enterprise frameworks.

03 Learn to work with AI.

04 Build a professional portfolio.

05 Prepare for the developer job market.

PREREQUISITE

Basic computer skills and a genuine interest in coding. The curriculum starts from Java fundamentals.

HOW YOU LEARN

Learn, practise, build and review — every module ends in practical work.

THE IDEAL LEARNER

Who is this programme for?

This programme is suitable for students who:

- Want to become professional software developers
- Are ready to learn Java from the fundamentals upward
- Are interested in enterprise development with Spring Boot
- Want both backend and frontend development skills
- Want to build real applications rather than only study theory
- Want a GitHub portfolio and interview preparation alongside training

PLEASE NOTE

You do not need prior professional experience. You do need commitment: 2 hours of focused training every day for 135 days, plus practice.

THE LEARNING LOGIC

Why Full Stack Java?

01 JAVA FOUNDATION

02 OBJECT-ORIENTED DESIGN

03 DATABASES & SQL

04 SPRING BOOT & SECURITY

05 FRONTEND & REACT

06 AI INTEGRATION

07 CAREER READINESS

Java and Spring Boot power a large share of enterprise software. This programme first builds strong Java and OOP foundations, then the frameworks companies use — and then how to use AI to build faster and add AI features to real applications.

The goal is not simply to use AI tools.
Engineer real software. Use AI to build it better.

WHAT YOU WILL WORK WITH

Technology stack

FRONTEND

HTML5 · CSS3 · Bootstrap · JavaScript · React

BACKEND

Java · JDBC · JSP · Spring · Hibernate · Spring Boot · Spring Security · JWT

DATABASE

MySQL · SQL

DEVELOPMENT TOOLS

Git · GitHub · Maven · Spring Initializr · Eclipse / IntelliJ · Postman

AI IN DEVELOPMENT

AI introduction · Vibe coding (AI-assisted development) · AI API integration · n8n workflow automation

GOOD TO KNOW

Spring Boot is the core backend framework.

Students first learn the layers underneath — **JDBC, JSP, Spring and Hibernate** — then build with **Spring Boot** and secure it with **Spring Security**.

React is part of the core curriculum (20 days).

12 PHASES · 135 DAYS

Your programme journey



Day 1 | Day 71 · Milestone Project 1 | Day 135 · Program end

01	ORIENTATION & INTRODUCTION	Days 1–2	4 Hrs
02	CORE JAVA & OOP	Days 3–42	80 Hrs
03	SQL & DATABASE ENGINEERING	Days 43–47	10 Hrs
04	ADVANCED JAVA & FRAMEWORKS	Days 48–64	34 Hrs
05	AI INTRODUCTION & VIBE CODING	Days 65–67	6 Hrs
06	MINI PROJECT (SPRING BOOT)	Days 68–71	8 Hrs
07	FRONTEND FOUNDATION: HTML, CSS & JAVASCRIPT	Days 72–95	48 Hrs
08	REACT	Days 96–115	40 Hrs
09	GITHUB & PORTFOLIO	Days 116–117	4 Hrs
10	AI-INTEGRATED MAIN PROJECT	Days 118–127	20 Hrs
11	AI AUTOMATION WITH N8N	Days 128–130	6 Hrs
12	BUFFER DAYS & FINAL REVIEW	Days 131–135	10 Hrs

TOTAL **135 Days** **270 Technical Hours**

Highlighted phases end with a major project. Your batch follows either a backend-first or a frontend-first order — the content and total hours are the same.

PHASE 01

01 Orientation & Introduction

Days 1–2 | **4** HOURS

01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12

Day 1 · Day 71 · Milestone Project 1 · Day 135 · Program end

WHAT YOU WILL LEARN

You will understand how the programme works, what you will build and how the learning route is organised.

KEY TOPICS

- Programme orientation
- Learning route
- Technical orientation
- Course introduction

PRACTICAL OUTCOMES

01 Clear programme roadmap

02 Understanding of the learning route

PHASE 02

02 Core Java & OOP

Days 3–42 | **80** HOURS

01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12

Day 1 · Day 71 · Milestone Project 1 · Day 135 · Program end

WHAT YOU WILL LEARN

You will learn Java from how it runs on the JVM to OOP, design patterns, exceptions, multithreading, Collections, Generics and Streams.

KEY TOPICS

Features of Java · JDK, JRE & JVM · Internal working · JDK install · Sample program · Packages · Class & object · Constructors · Access specifiers · Data types · Type casting · Methods · Call by value / reference · Operators · Scanner class · Conditional statements · Loops · Switch · Encapsulation · Inheritance · Abstraction · Polymorphism · Composition over inheritance · Constructor chaining · Arrays & types · For-each · Anonymous arrays · Advanced array programs · String methods · StringBuffer & StringBuilder · String practice (Day 23) · Singleton · Strategy · Factory · OOP + design pattern application · Exception types · Keywords · Nested exceptions · Custom exceptions · Thread creation · Thread lifecycle · Synchronisation · Inter-thread communication · Instance vs local variables · List & Queue · Set & Map · Generics · Wildcards · Functional interfaces · Lambda expressions · Comparable & Comparator · Stream API · Optional

PRACTICAL OUTCOMES

01 Java programming exercises

02 OOP and design-pattern applications

03 Multithreaded programs

04 Collections & Stream API programs

PHASE 03

03 SQL & Database Engineering

Days 43–47 | **10** HOURS



Day 1 | Day 71 · Milestone Project 1 | Day 135 · Program end

WHAT YOU WILL LEARN

You will design databases, apply constraints and write queries — from basic commands to joins and normalisation.

KEY TOPICS

Data integrity | SQL operators | Database creation | SQL constraints | SQL commands | Data manipulation | Aggregate functions | ORDER BY | GROUP BY | ALTER TABLE | Wildcard characters | JOINS | Normalisation | SQL practice

PRACTICAL OUTCOMES

01 Database schema

02 SQL exercise set

03 Multi-table queries

04 Normalised design

PHASE 04

04 Advanced Java & Frameworks

Days 48–64 | **34** HOURS

01 | 02 | 03 | **04** | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12

Day 1 — Day 71 · Milestone Project 1 — Day 135 · Program end

WHAT YOU WILL LEARN

You will connect Java to MySQL, build web pages with JSP, use Spring and Hibernate, and build secure applications with Spring Boot and Spring Security.

KEY TOPICS

JDBC | JDBC CRUD | JSP | JSP CRUD | Spring | Hibernate | Hibernate CRUD | Spring Initializr | Mapping | Spring Boot CRUD | Annotations | Security concepts | JWT authentication | Security filter chain | Role-protected endpoints

PRACTICAL OUTCOMES

01 JDBC CRUD application

02 JSP web application

03 Spring Boot CRUD API

04 JWT-secured Spring Boot API

PHASE 05

05 AI Introduction & Vibe Coding

Days 65–67 | **6** HOURS



WHAT YOU WILL LEARN

You will learn the basics of AI and how to use AI-assisted coding tools productively — without losing understanding of your own code.

KEY TOPICS

- AI fundamentals
- AI in software development
- AI-assisted coding
- Prompting for code

PRACTICAL OUTCOMES

01 AI concepts overview

02 AI-assisted build exercise

PHASE 06

06 Mini Project (Spring Boot)

Days 68–71 | **8** HOURS

01 | 02 | 03 | 04 | 05 | **06** | 07 | 08 | 09 | 10 | 11 | 12

Day 1 · Day 71 · Milestone Project 1 · Day 135 · Program end

WHAT YOU WILL LEARN

You will build a Spring Boot mini project that brings together everything from the backend phases.

KEY TOPICS

Project scope · Database design · API planning · Java · MySQL · Spring Boot · Hibernate · Spring Security · JWT · Mock interview (Day 71)

PRACTICAL OUTCOMES

01 Approved project plan

02 Working Spring Boot mini project

03 Mock interview experience

PHASE 07

07 Frontend Foundation: HTML, CSS & JavaScript

Days 72–95 | **48** HOURS

01 | 02 | 03 | 04 | 05 | 06 | **07** | 08 | 09 | 10 | 11 | 12

Day 1 — Day 71 · Milestone Project 1 — Day 135 · Program end

WHAT YOU WILL LEARN

You will learn how web pages are structured, styled and made interactive with JavaScript.

KEY TOPICS

HTML structure · Text formatting · CSS types & properties · Tables & lists · HTML forms · Position properties · Flexbox · Grid · Semantic elements · Bootstrap · Webpage build · Variables · Data types & operators · Type coercion · Conditionals & switch · Loops · Functions · Strings & arrays · Objects · DOM · Events · JS OOP · Asynchronous functions · Error handling · JSON data handling

PRACTICAL OUTCOMES

- **01** Static website
- **02** Responsive Bootstrap site
- **03** Interactive JavaScript pages
- **04** Async JSON application

PHASE 08

08 React

Days 96–115 | **40** HOURS

01 | 02 | 03 | 04 | 05 | 06 | 07 | **08** | 09 | 10 | 11 | 12

Day 1 | Day 71 · Milestone Project 1 | Day 135 · Program end

WHAT YOU WILL LEARN

You will build modern React applications — components, hooks, routing, data fetching and CRUD.

KEY TOPICS

React intro · Components & JSX · Props & state · Event handling · Styling in React · React in depth · Rendering & lifecycle · useState · useEffect · useRef · Custom hooks · Conditional rendering · Lists · React forms · Prop drilling & useContext · React Router · useNavigate · useLocation · Lazy loading · Error boundaries · Data fetching · Local storage · CRUD · CRUD with local storage

PRACTICAL OUTCOMES

01 React component app

02 Routed single-page app

03 React CRUD application

PHASE 09

09 GitHub & Portfolio

Days 116–117 | **4** HOURS



WHAT YOU WILL LEARN

You will put your projects on GitHub and organise them as a professional portfolio.

KEY TOPICS

- Version control
- Repositories
- Portfolio organisation

PRACTICAL OUTCOME

01 GitHub portfolio

PHASE 10

10 AI-Integrated Main Project

Days 118–127 | **20** HOURS



Day 1 | Day 71 · Milestone Project 1 | Day 135 · Program end

WHAT YOU WILL LEARN

You will build your main project: a Java full-stack application with AI features integrated through APIs.

KEY TOPICS

Spring Boot | Spring Security | MySQL | JWT | React / JavaScript | AI API integration | GitHub

PRACTICAL OUTCOMES

01 Java full-stack AI-integrated application

02 Project on GitHub

PHASE 11

11 AI Automation with n8n

Days 128–130 | **6 HOURS**



Day 1 | Day 71 · Milestone Project 1 | Day 135 · Program end

WHAT YOU WILL LEARN

You will build automated workflows that connect apps and AI using n8n.

KEY TOPICS

n8n | Workflow automation | AI in workflows

PRACTICAL OUTCOME

01 Automated AI workflow

PHASE 12

12 Buffer Days & Final Review

Days 131–135 | **10** HOURS



Day 1 | Day 71 · Milestone Project 1 | Day 135 · Program end

WHAT YOU WILL LEARN

Reserved time for catch-up, finishing your projects and revision.

KEY TOPICS

- Catch-up
- Project completion

FINAL OUTCOMES

01 Completed portfolio

02 Final review

DEMONSTRABLE PROJECT WORK

Two major projects

Days 68–71

MILESTONE PROJECT 1

Spring Boot Mini Project

YOU DEMONSTRATE

Database design · CRUD · REST endpoints · Persistence · Authentication · Role-protected endpoints · Java · MySQL · JDBC · Hibernate

Days 118–127

MAIN PROJECT

API-Integrated Full-Stack Application with AI

YOU DEMONSTRATE

Backend API · Database · Authentication · React / JS frontend · API integration · AI features · Code on GitHub · Spring Boot · Spring Security · MySQL · JWT

Students do not finish with only certificates.
They finish with demonstrable project work.

CHECKING YOUR PROGRESS

Reviews & interview readiness

Your progress is reviewed at the end of every major module, and you will face a mock interview before placement.

CHECKPOINT	WHEN	COVERS
Internal review	**Day 26**	Java basics, OOP, arrays & strings
Internal review	**Day 42**	Advanced core Java (including SQL)
Internal review	**Day 64**	JDBC, JSP, Spring, Hibernate & Spring Boot
Mock interview	**Day 71**	Right after the mini project
Review	**Day 95**	HTML, CSS & JavaScript
Review	**Day 115**	React
Final review	**Day 135**	Whole programme & portfolio

REVIEWS

Trainers check your understanding, your code and your project progress — and tell you exactly what to improve.

MOCK INTERVIEW

Practise a real technical and HR-style interview before you apply for jobs.

PLACEMENT SUPPORT AND CAREER ASSISTANCE

Kompetenzen digital career & placement ecosystem

As a student, you can access the Kompetenzen digital placement ecosystem at www.kompetenzen.in for:

01
Job discovery
Students can explore job opportunities listed on the Kompetenzen platform.

02
Direct application
Students can apply directly to suitable job opportunities.

03
ATS resume
Students can create and maintain an ATS-friendly resume.

04
Career profile
Students can maintain their professional profile for placement opportunities.

05
Consultant support
Students can contact Kompetenzen career/placement consultants for guidance and assistance.

06
Interview readiness
Reviews and the mock interview prepare students for applications and interviews.

These are placement support and career assistance services. They do not constitute a guarantee of employment.

YOUR PORTFOLIO PROGRESSION

What you will be able to build

- 01 JAVA PROGRAMS & OOP APPLICATION
- 02 DESIGN-PATTERN APPLICATION
- 03 COLLECTIONS & STREAMS PROGRAMS
- 04 JDBC CRUD APPLICATION
- 05 JSP WEB APPLICATION
- 06 JWT-SECURED SPRING BOOT API
- 07 SPRING BOOT MINI PROJECT
- 08 INTERACTIVE JAVASCRIPT APP
- 09 REACT CRUD APPLICATION
- **10 AI-INTEGRATED MAIN PROJECT**

FINAL STUDENT OUTCOME

> By the end of the programme, students will have developed the skills to build full-stack applications using Java, Spring Boot and React, work with databases through JDBC and Hibernate, secure APIs with Spring Security and JWT, use AI-assisted development, integrate AI into applications, automate workflows with n8n and present their work professionally.

Engineer. **Build.** **Integrate.** **Present.**